UE26MA141A Engineering Mathematics – I · Unit 1 · Partial Differentiation

Graphs and diagrams

Every graph and diagram in scope, redrawn as clean vector figures.

Class 002 · Geometrical interpretation of partial derivatives · compiled 14 October 2026

ESA ★★★	Asked repeatedly in past ESA papers
ESA ★★	Asked in past ESA papers or computed in class-work problems
ESA ★	Conceptual: taught in class, not examined so far
ADDED FOR CLARITY	Standard context that is not in the class material
TRANSCRIPT ONLY	Said in the lecture but not on any slide
VIDEO ONLY	Seen only in a frame of the class video
NOT ON CURRENT SLIDES	From older material or past papers, not this year's slides
S002·7	Deck 002, slide 7 (rendered .pptx)
T002 00:06:10	AV-summary transcript 002, segment start time
V002 00:07:45	Frame grabbed from the class 002 video
CW U1 P2 · ESA Dec-23 1a	Unit 1 class-work problem 2 · past-paper question

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Colours follow the Okabe–Ito order: blue = main curve, vermilion = comparison or tangent, green = second trace, purple = other direction. Tangents are dashed. Grey dashed lines are references. The surfaces (dome $z = 4 - x^2 - y^2$, saddle $z = x^2 - y^2$, $z = x^2 y$) are illustrative choices that reproduce the shapes on slides 5–10. The slide images themselves are third-party screenshots and are not copied.

2 The one-variable picture: derivative as slope

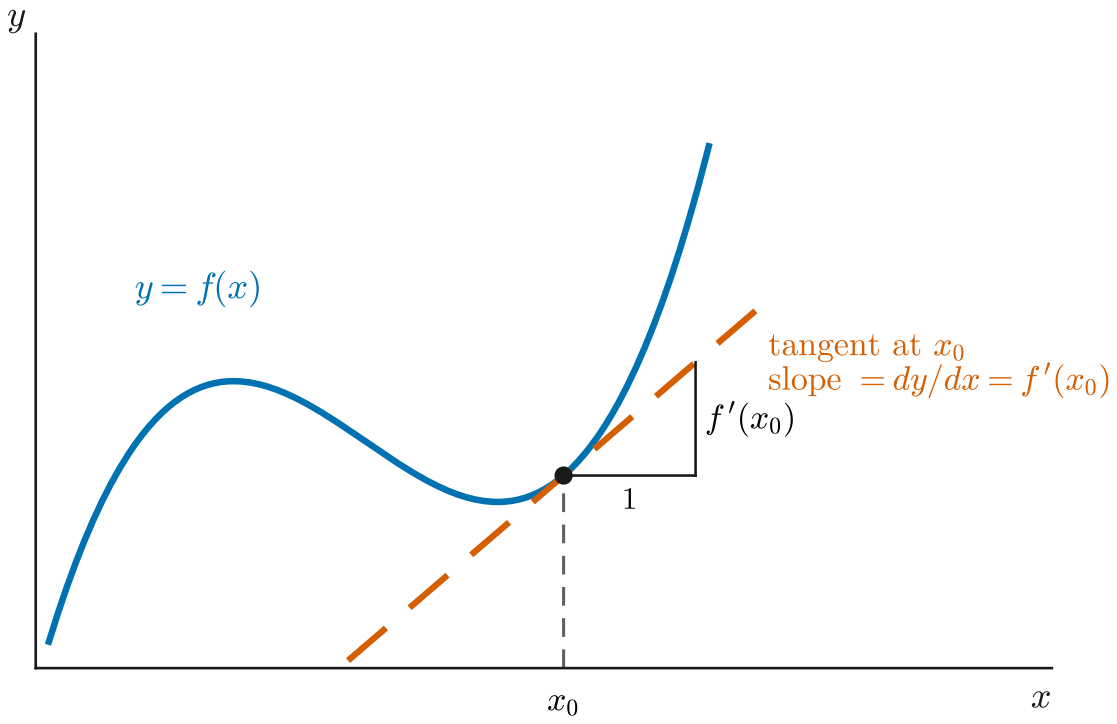


Figure G2.1. Derivative of y = f(x) as the slope of the tangent. ESA ★

S002·4 · T002 00:03:13

How to read it. The blue curve is $y = f(x)$, and the dashed vermilion line is its tangent at x_0. The small triangle shows the slope: a run of 1 gives a rise of $f'(x_0)$. In an exam, label the point, the tangent and the slope dy/dx at x_0.

3 A function of two variables is a surface

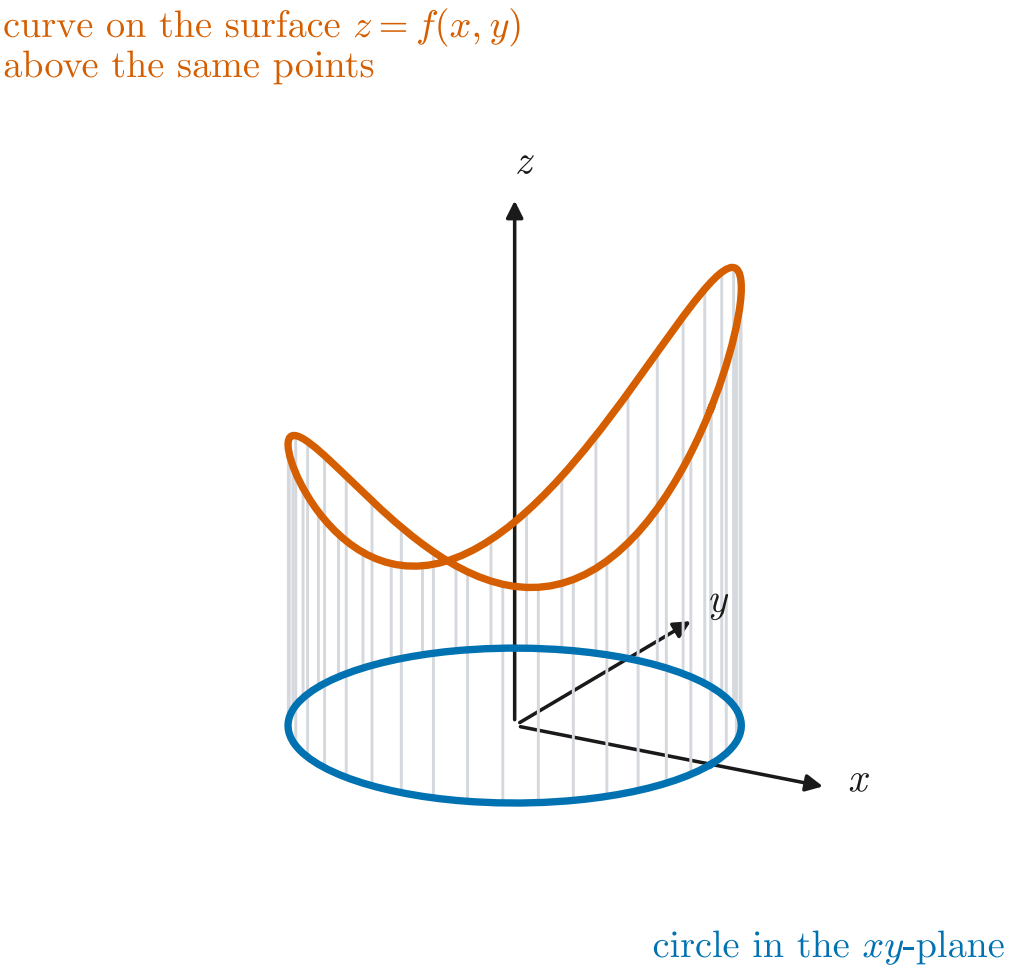


Figure G3.1. Points on a circle in the xy-plane give a curve on the surface. ESA ★

S002·5 · T002 00:03:32

How to read it. The blue circle lies in the xy-plane. Each grey line rises to $z = f(x, y)$, and the vermilion curve joins the tops: a closed curve lying on the surface $z = f(x, y) = 2.2 + 0.8xy + 0.5x$ (an illustrative choice).

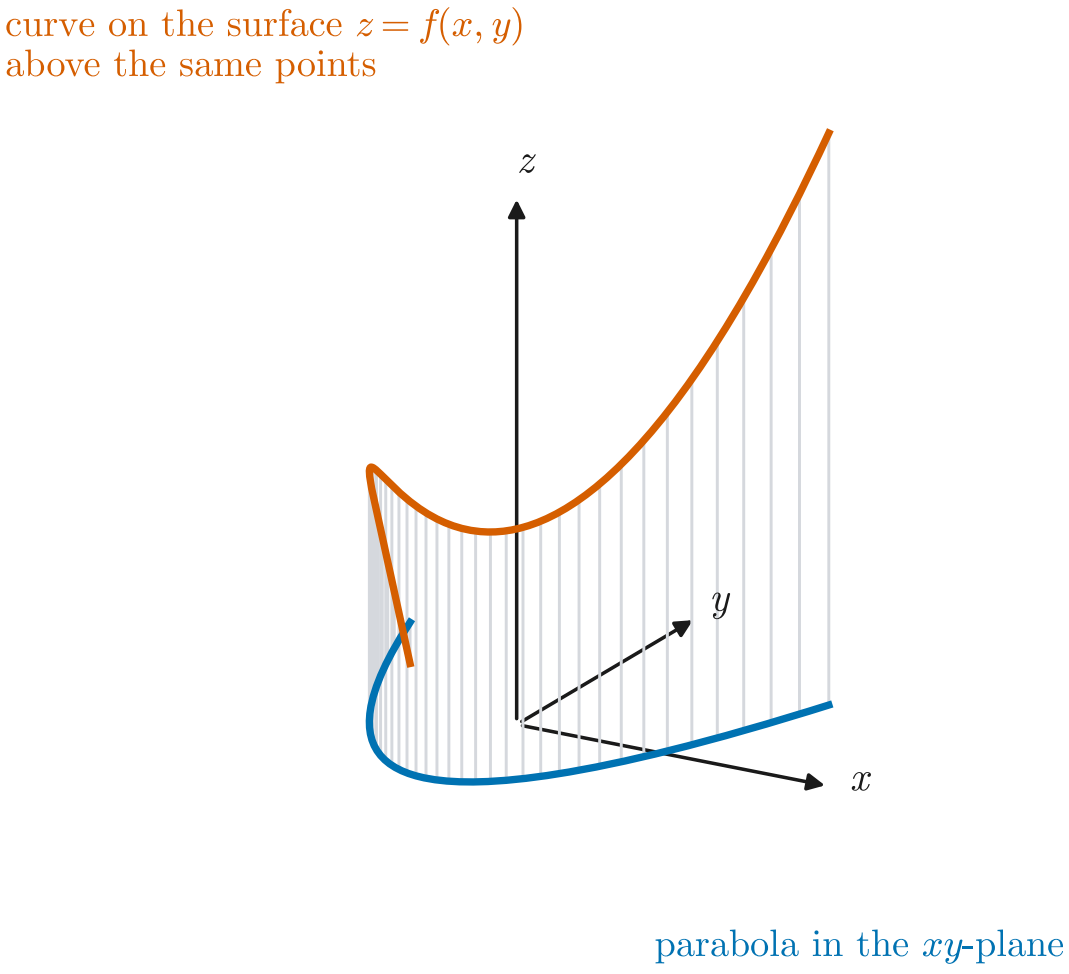


Figure G3.2. Points on a parabola in the xy-plane give a curve on the surface. ESA ★

S002·5 · T002 00:03:47

How to read it. Same idea with a parabola $y = x^2 - 1.2$ in the xy-plane. Every point of the parabola has one height on the surface, so the vermilion curve lies on the surface above it.

4 First-order partials as slopes of tangents

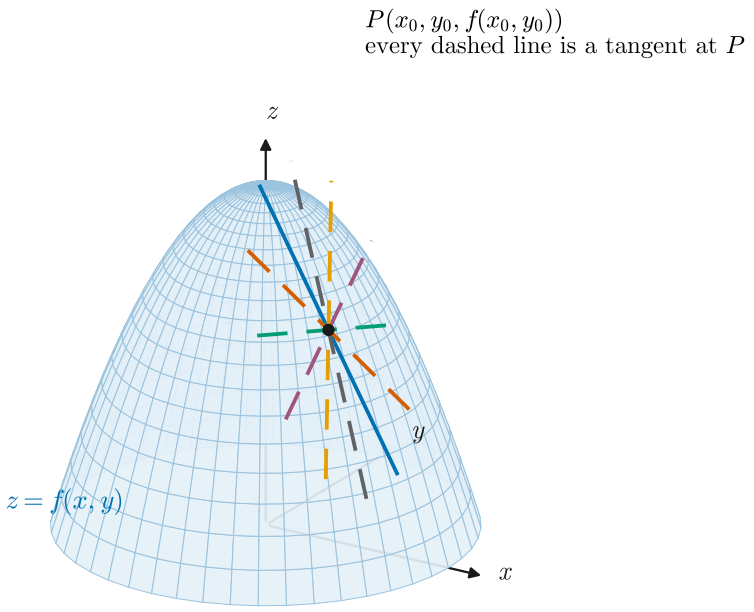


Figure G4.1. A surface has infinitely many tangents at a point. ESA ★★

S002·6 · T002 00:04:36 · T002 00:05:24

How to read it. The dome is $z = 4 - x^2 - y^2$ and $P = (1, -0.5, 2.75)$. Each coloured line is a tangent to the surface at P in a different direction. Partial differentiation picks out two of them: the ones parallel to the x- and y-axes.

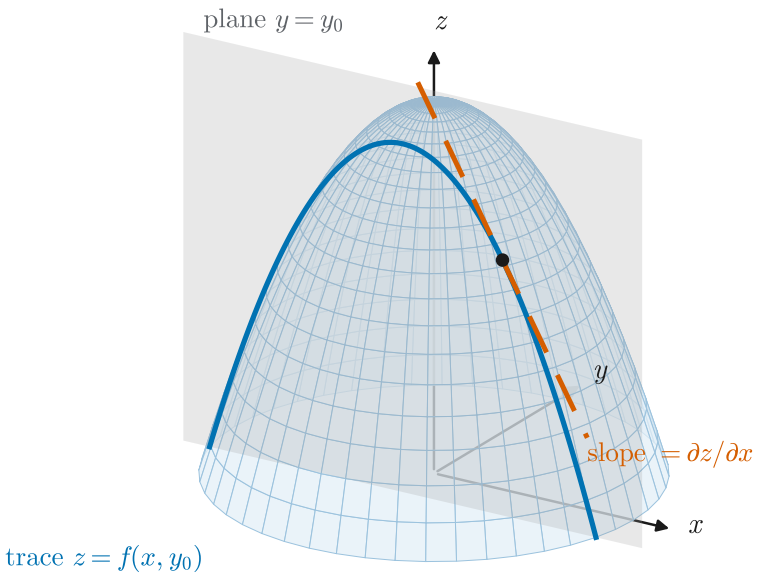


Figure G4.2. $\partial z/\partial x$: the tangent parallel to the x-axis. ESA ★★

S002·7 · V002 00:07:45 · T002 00:06:54

How to read it. The grey plane $y = y_0$ cuts the dome in the blue trace $z = f(x, y_0)$. The dashed tangent to that trace at P is parallel to the x-axis, and its slope is $\partial z/\partial x$ at (x_0, y_0).

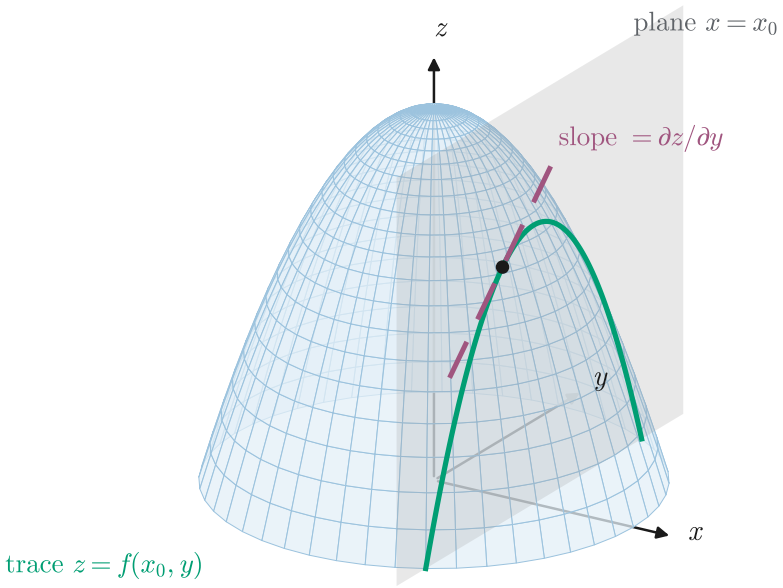


Figure G4.3. $\partial z/\partial y$: the tangent parallel to the y-axis. ESA ★★

S002·7 · V002 00:07:45 · T002 00:07:10

How to read it. The plane $x = x_0$ cuts the dome in the green trace $z = f(x_0, y)$, which "just depends on y". The dashed tangent at P is parallel to the y-axis, and its slope is $\partial z/\partial y$ at (x_0, y_0).

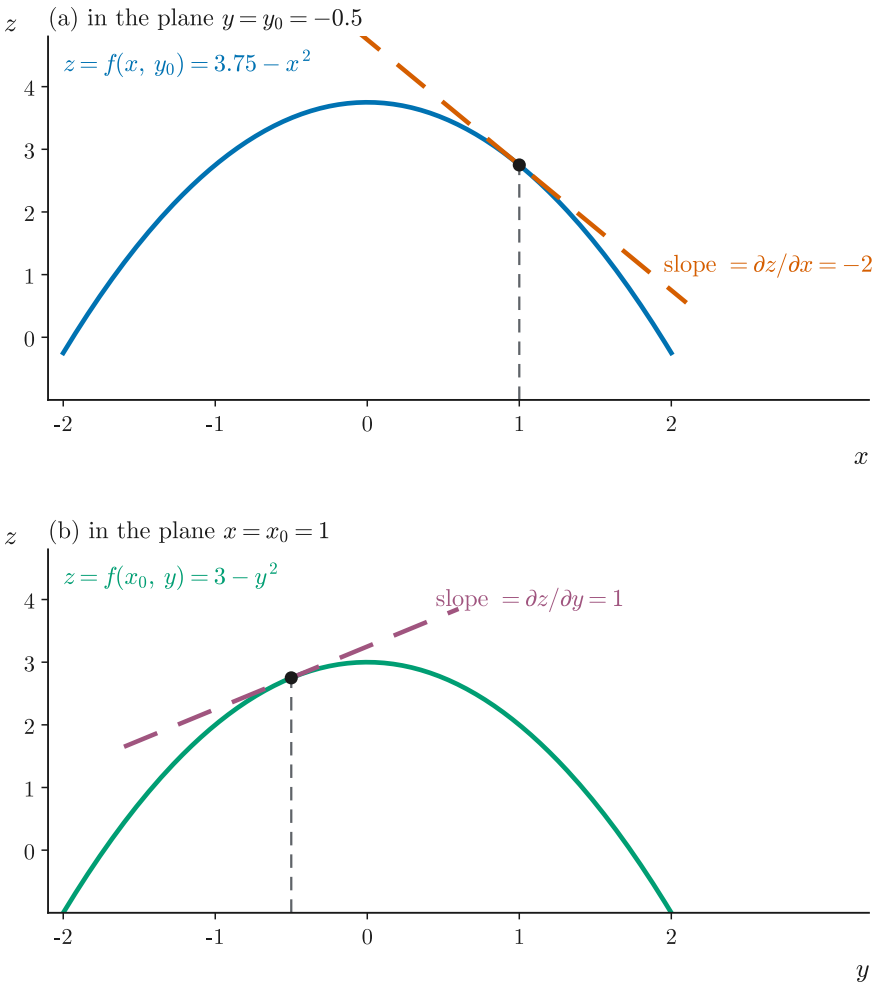


Figure G4.4. The two traces seen flat, with numerical slopes. ESA ★★

ADDED FOR CLARITY S002·7

How to read it. These are panels (a) and (b) from Figures G4.2 and G4.3, drawn in their own planes. At P, the tangent in (a) has slope $-2 = f_x(1, -0.5)$ and the tangent in (b) has slope $1 = f_y(1, -0.5)$. See worked example W4.1.

5 Slope in any other direction: the gradient

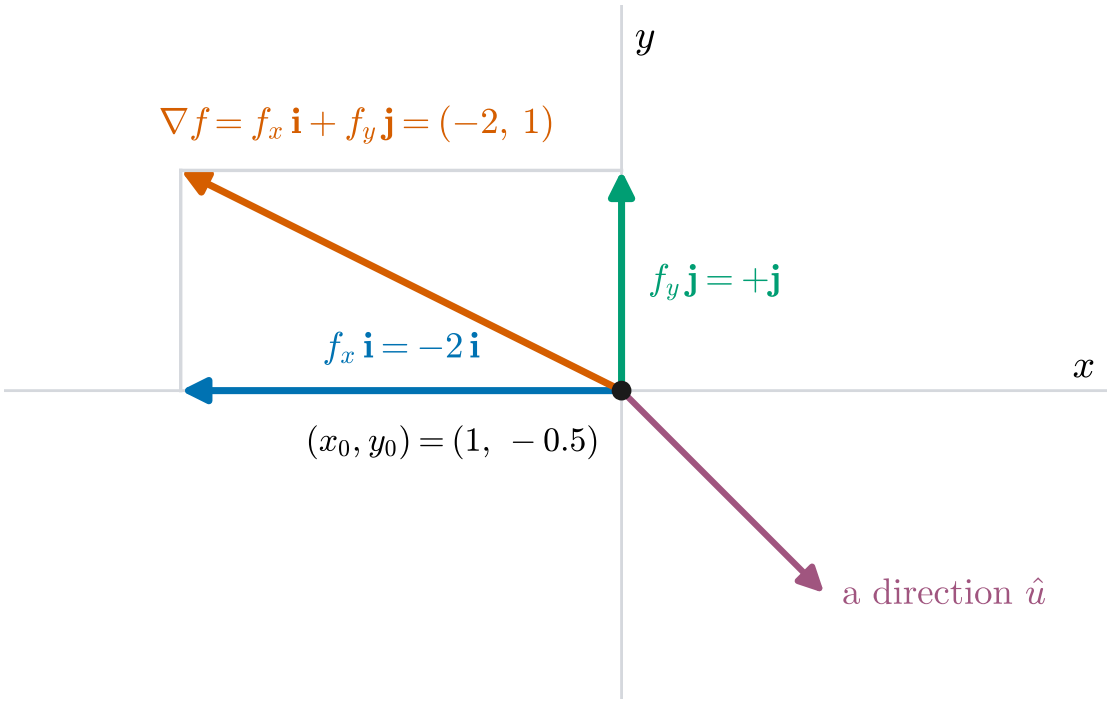


Figure G5.1. Resolving into components: the gradient. ESA ★

S002·8 · T002 00:09:06 · V002 00:09:05

How to read it. The slope components $f_x\,\mathbf{i}$ (blue) and $f_y\,\mathbf{j}$ (green) add by vector addition to give the resultant ∇f (vermilion), drawn here for the dome at $(1, -0.5)$. The purple arrow is another direction $\hat{u}$; the slope along it is $\nabla f \cdot \hat{u}$ (W5.1).

6 Second-order partials: concavity and mixed partials

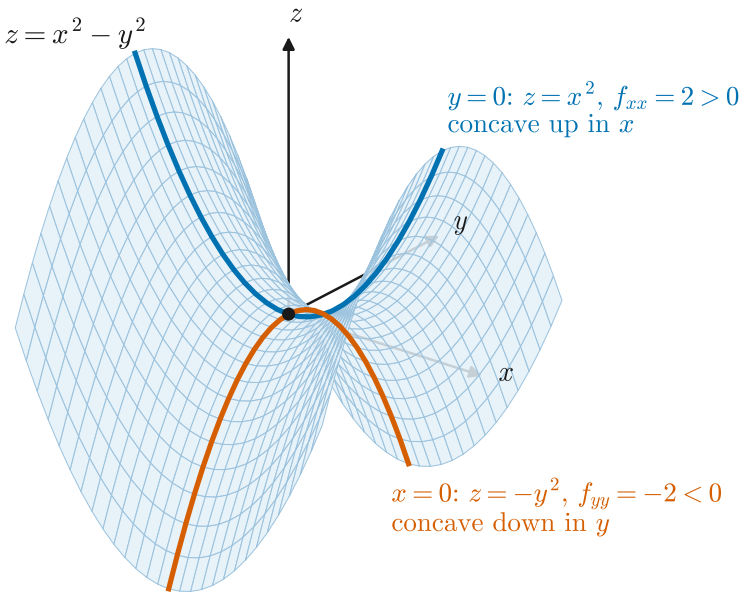


Figure G6.1. Concave up in x, concave down in y: the saddle z = x² − y². ESA ★★

ADDED FOR CLARITY S002·9 · S002·10 · T002 00:11:19

How to read it. Along the x direction (blue trace, $y = 0$) the surface bends up, so $f_{xx} = 2 > 0$. Along the y direction (vermilion trace, $x = 0$) it bends down, so $f_{yy} = -2 < 0$. One point can be concave up in one direction and concave down in another.

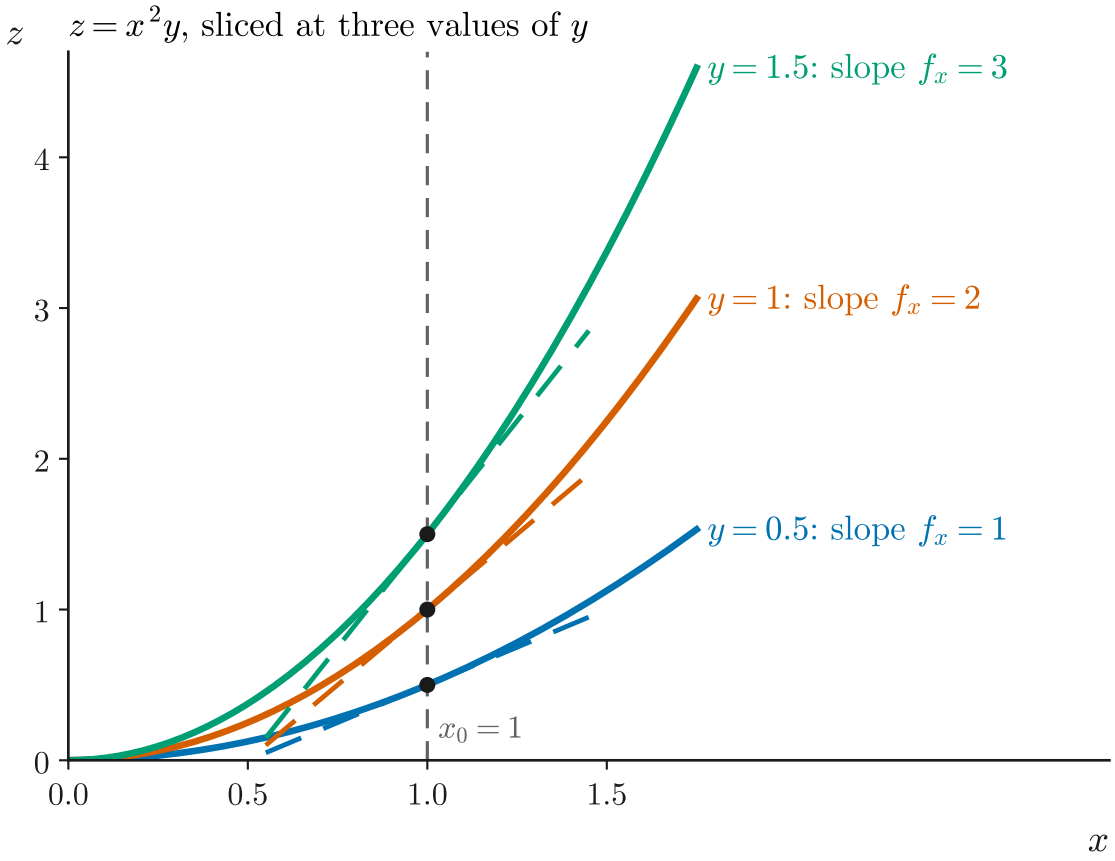


Figure G6.2. Mixed partial: how the x-slope changes as y changes. ESA ★★

ADDED FOR CLARITY S002·9 · T002 00:11:45

How to read it. Three traces of $z = x^2 y$, at $y = 0.5$, 1 and 1.5. At $x_0 = 1$ their tangents have slopes $f_x = 1, 2, 3$. The x-slope grows by 2 per unit of y, which is $f_{xy} = 2$.